Gopal Krishna JS

MCA Student
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Portfolio

Summary

MCA student at RV College of Engineering with experience building scalable backend systems, real-time analytics platforms, and AI-powered applications, with interests in Linux systems and performance engineering.

Work Experience

- **COE Intern — AI Research and Business Solutions** *Aug 2026 – Dec 2026*
  Center of Excellence, RVCE–Boston *Hybrid*
  - Analyzed three years of Sentinel-2 satellite imagery across Sringeri and Sulya, covering 72 monthly datasets at 10 m spatial resolution.
  - Computed 15 vegetation indices and applied Random Forest classification to map arecanut plantations and categorize crop health as Healthy, Stressed, and Suspected YLD.
  - Integrated satellite imagery, weather data, and GIS layers to identify disease hotspots and support early detection, in collaboration with RRSC-South and CPCRI.

Projects

- **SkyPulse — Global Flight Heatmap Analytics Platform** *2026 – Present*
  React, TypeScript, FastAPI, PostgreSQL/PostGIS, Redis, Docker, WebSockets **GitHub**
  - Engineered a real-time geospatial analytics platform to visualize 1,000+ aircraft records on interactive Leaflet maps using WebSocket streaming and dynamic heatmaps.
  - Optimized real-time data delivery through Redis caching, database indexing, and geospatial query optimization, achieving approximately 500 ms latency under tested conditions.
- **Veles** *Jul 2026 – Sep 2026*
  Python, C++, REST APIs, Docker, AWS **GitHub**
  - Engineered a modular backend system using Python and C++ to automate resource management and streamline data-processing workflows.
  - Integrated AWS S3 object storage for scalable file storage and retrieval, enabling centralized management of application data and artifacts.
  - Developed reusable utility modules and automated data pipelines, leveraging Docker for consistent execution environments and reproducible deployments.

Technical Skills

Languages: C++, Rust, Python, TypeScript, SQL, Bash
Backend & Databases: FastAPI, Node.js, REST APIs, PostgreSQL, PostGIS, Redis
Systems & Infrastructure: Linux, AWS, Docker, Kubernetes, Git, GNS3
Libraries & Tools: NumPy, Pandas, Matplotlib, Unity 3D, Figma, Adobe Creative Suite

Achievements

- **2nd Place** — ABBS Bengaluru Hackathon 2026 (Aegis — Real-Time Phishing and Fraud Detection Platform).
- **Participant** — FantomCode, RVITM Bengaluru Hackathon 2026 (OpenAudit).

Education

- **RV College of Engineering** *2025 – 2027 (Expected)*
  Master of Computer Applications *CGPA: 8.48/10*
- **Sri Bhuvanendra College, Karkala** *2022 – 2025*
  Bachelor of Computer Applications *CGPA: 9.00/10*

Positions of Responsibility

- **Student Placement Coordinator — RVCE** *Apr 2026 – Present*
  RV College of Engineering, Bangalore
  - Coordinate between recruiters and students for campus placements.
- **Student Coordinator — ICOSICS 2026** *Sep 2026*
  RV College of Engineering, Bangalore
  - Organized and coordinated conference proceedings, speaker management, and event logistics for ICOSICS 2026, an IEEE conference focused on Secure IoT and Cybersecurity.
- **Student Coordinator — IEEE Workshop on Gen AI and Agentic Intelligence** *Apr 2026*
  RV College of Engineering, Bangalore
  - Organized and coordinated workshop activities, speaker engagement, and event logistics focused on Generative AI and Agentic Intelligence under IEEE.